# Project SOA:
# Theory of Shared-Basis Attention for BERT-base on Apple M1 Hardware

Research Dossier

**Abstract**

This paper presents a theoretical framework and methodology for *Project SOA*, which proposes replacing classical Query/Key/Value (Q/K/V) projection matrices in BERT-base attention layers with a *shared global orthogonal basis* constructed via Singular Value Decomposition (SVD). This approach drastically reduces parameter count and computational cost, enabling efficient long-context transformer inference on Apple M1's Unified Memory Architecture and Neural Engine hardware. We provide a comprehensive linear algebraic justification, subspace alignment theory across layers, optimization implications of a fixed shared basis, and layout considerations for rank selection and hardware deployment. This concise paper targets a compact formal theory presentation supplemented by empirical methodology plans.

## Contents

## 1 Problem Setup and Notation

Consider BERT-base architecture with:

- Number of layers: $L = 12$
- Hidden dimension: $d = 768$
- Number of attention heads: $h = 12$, head size $d_h = 64$
- Input sequence length $T$ (targeted for long contexts $T = 2k$ to $4k$)

At each layer $\ell \in \{1, \ldots, L\}$, define the standard attention linear projections:

$$W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)} \in \mathbb{R}^{d \times d}.$$

Given input hidden states $X^{(\ell)} \in \mathbb{R}^{T \times d}$,

$$Q^{(\ell)} = X^{(\ell)} W_Q^{(\ell)}, \quad K^{(\ell)} = X^{(\ell)} W_K^{(\ell)}, \quad V^{(\ell)} = X^{(\ell)} W_V^{(\ell)}.$$

Multi-head attention is fused conceptually into single Q/K/V mappings per layer.

### 1.1 Shared-Basis Parameterization

Project SOA replaces each attention weight matrix by:

$$W_*^{(\ell)} \approx U A_*^{(\ell)},$$

where:

- $U \in \mathbb{R}^{d \times r}$ is a global orthonormal basis with rank $r \ll d$
- $A_*^{(\ell)} \in \mathbb{R}^{r \times d}$ are small trainable projection matrices

Here, $U$ is fixed (frozen) after estimation, shared across all Q/K/V and all layers, and $A_*^{(\ell)}$ are adapted in training or derived from pretrained weights.

Two main constructions for $U$ are considered:

1. **Pretrained-Weight SVD:** SVD on concatenated $W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)}$
2. **Covariance-Based SVD:** SVD on hidden-state covariance matrix across layers/positions

## 2 Numerical Linear Algebra Foundations

### 2.1 Eckart–Young–Mirsky Theorem

**Theorem 2.1** (Eckart–Young–Mirsky)**.** *Let matrix $W \in \mathbb{R}^{d \times d}$ have singular value decomposition*

$$W = U_W \Sigma_W V_W^\top,$$

*with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_d \geq 0$.*

*For any $r < d$, the best rank-$r$ approximation to $W$ minimizing the Frobenius norm error is*

$$W_r := U_W^{(:,1:r)} \Sigma_W^{(1:r,1:r)} V_W^{(:,1:r)\top},$$

*and*

$$\min_{\mathrm{rank}(Z) \leq r} \|W - Z\|_F = \left( \sum_{i>r} \sigma_i^2 \right)^{1/2} = \|W - W_r\|_F.$$

*Sketch.* See standard linear algebra references; this follows from properties of SVD truncation preserving optimality in unitarily invariant norms. □

This theorem justifies factorizing attention projections as $W \approx UA$, where $U = U_W^{(:,1:r)}$.

### 2.2 Shared-Basis Approximation Across Multiple Matrices

Let $\{W^{(m)}\}_{m=1}^M$ denote all $3L$ attention matrices. We seek a single orthonormal basis $U \in \mathbb{R}^{d \times r}$ minimizing the total approximation error:

$$\min_{U^\top U = I_r} \sum_{m=1}^{M} \min_{A^{(m)}} \|W^{(m)} - U A^{(m)}\|_F^2.$$

**Lemma 2.2** (Optimal $A^{(m)}$ for fixed $U$)**.** *Given fixed $U$, the minimizer is $A^{(m)*} = U^\top W^{(m)}$ with residual error*

$$\|W^{(m)} - U A^{(m)*}\|_F^2 = \|(I - UU^\top) W^{(m)}\|_F^2.$$

*Proof.* By orthogonal projection properties and Pythagorean theorem. □

Therefore, the problem reduces to:

$$\min_{U^\top U = I_r} \sum_{m=1}^{M} \|(I - UU^\top)W^{(m)}\|_F^2.$$

Stacking all column vectors of $W^{(m)}$ into $\mathcal{X} \in \mathbb{R}^{d \times N}$, where $N = M \times d$, this is the classical PCA objective on $\mathcal{X}$.

**Proposition 2.3** (Shared-Basis PCA)**.** *The minimizing $U$ consists of the top $r$ left singular vectors of $\mathcal{X}$.*

*Sketch.* Equivalent to projection onto best rank $r$ subspace minimizing reconstruction error. □

### 2.3 Global Energy Criterion for Rank Selection

Define total energy

$$E_{\text{total}} = \|\mathcal{X}\|_F^2 = \sum_i \sigma_i^2(\mathcal{X}),$$

and rank-$r$ preserved energy

$$E(r) = \sum_{i=1}^{r} \sigma_i^2(\mathcal{X}).$$

Choose rank $r$ s.t.

$$\eta(r) := \frac{E(r)}{E_{\text{total}}} \geq \tau,$$

for threshold $\tau \approx 0.99$.

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## 3 Subspace Alignment and Shared Basis Justification

**Definition 3.1** (Subspace Distance)**.** For two $r$-dimensional orthonormal bases $U^{(1)}, U^{(2)} \in \mathbb{R}^{d \times r}$, define

$$d_{\text{sub}}(U^{(1)}, U^{(2)}) := \max_{i=1}^{r} \sin\theta_i,$$

where $\{\theta_i\}$ are principal angles between the two subspaces.

**Proposition 3.1** (Error bound with shared basis)**.** *If $W_r^{(m)}$ is the best rank-$r$ approximation of $W^{(m)}$ with subspace $\mathcal{S}^{(m)}$, and $U$ is a shared basis, then*

$$\|W^{(m)} - UA^{(m)}\|_F^2 \leq \|W^{(m)} - W_r^{(m)}\|_F^2 + C_m d_{\text{sub}}^2(\mathcal{S}^{(m)}, \text{span}(U)),$$

*for some constant $C_m$.*

*Sketch.* Employ decomposition of error into components in and orthogonal to shared basis and apply Davis-Kahan-type bounds. □

This motivates the *Shared Latent Space Hypothesis*: the subspaces $\mathcal{S}^{(m)}$ are well-aligned across all $m$, justifying a single $U$.

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# 4 Shared Basis Construction Methods

## 4.1 Pretrained-Weights SVD

Define

$$\mathcal{X}_{\mathrm{w}} = \left[ W_Q^{(1)} \mid W_K^{(1)} \mid W_V^{(1)} \mid \cdots \mid W_V^{(L)} \right].$$

Calculate

$$\mathcal{X}_{\mathrm{w}} = U_{\mathrm{w}} \Sigma_{\mathrm{w}} V_{\mathrm{w}}^{\top},$$

and take $U = U_{\mathrm{w}}^{(:,1:r)}$.

Then set $A_*^{(\ell)} = U^{\top} W_*^{(\ell)}$.

## 4.2 Covariance-Based SVD

Estimate covariance

$$C = \frac{1}{N} \sum_{i,\ell} (x_i^{(\ell)} - \mu)(x_i^{(\ell)} - \mu)^{\top},$$

of hidden states, with eigendecomposition

$$C = U_{\mathrm{c}} \Lambda U_{\mathrm{c}}^{\top}.$$

Take $U = U_{\mathrm{c}}^{(:,1:r)}$.

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# 5 Optimization Theory

## 5.1 Gradient Flow through Fixed Basis

With

$$W = UA, \quad U^{\top} U = I_r,$$

gradients w.r.t. $A$ are

$$\frac{\partial L}{\partial A} = U^{\top} \frac{\partial L}{\partial W}.$$

Thus optimization acts as projected gradient descent onto the range of $U$.

## 5.2 Regularization and Generalization

The low-rank constraint reduces effective parameter count, induces implicit regularization, and improves optimization conditioning.

## 5.3 Migration vs From-Scratch Training

Migration uses pretrained decomposition to initialize $A$, facilitating fast convergence. From scratch involves learning $U, A$, often less stable and with higher compute.

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# 6 Rank Sweep and Hardware Complexity

## 6.1 Parameter Count Reduction

Original attention_params:

$$3Ld^2,$$

SOA attention_params:

$$dr + 3Lrd = dr(1 + 3L).$$

Relative ratio $\approx \frac{37r}{36d}$ for BERT-base ($L = 12, d = 768$), resulting in $4\% - 17\%$ reduction for ranks examined.

## 6.2 FLOPs Reduction

Original FLOPs per layer per projection $\approx 3Td^2$, SOA FLOPs

$$\approx 4Tdr,$$

achieving up to $\approx 22\%$ of original projection cost at largest tested rank.

## 6.3 M1/ANE Mapping

Shared $U$ enables reusing the large $d \times r$ matrix efficiently on ANE; $A_*^{(\ell)}$ matrices are small and easier to quantize and manage.

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# 7 Empirical Methodology (Outline)

- Construct $U$ using both pretrained-weights and covariance-based methods for ranks $\{32, 48, 64, 96, 128\}$.
- Initialize $A_*^{(\ell)}$ from regression or pretrained weights.
- Fine-tune only $A_*^{(\ell)}$ (optionally with other parameters).
- Evaluate GLUE/SQuAD tasks, long-context stability on M1, energy, latency.
- Measure subspace alignment using principal angles.

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# 8 Summary

Project SOA leverages classical numerical linear algebra and transformer theory to formulate a practical, theoretically well-founded approach to low-rank, shared-basis attention in BERT-base. The method is optimized for Apple M1 hardware enabling efficient long context attention. The fixed shared basis $U$ is justified through PCA and subspace alignment theory. Optimization occurs over low-dimensional factors $A_*^{(\ell)}$ with implicit projection, providing regularization and feasibility.

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## Acknowledgements

This dossier compiles theoretical insights synthesized to support Project SOA initiatives for M1-optimized transformers.